

Lloyds Banking Group

Customer Churn Prediction Model

Comprehensive Analysis Report

Algorithm · Performance Metrics · Business Strategy · Improvement Roadmap

5,204	17	99%	0.999
Records	Features	Accuracy	ROC-AUC

1. Executive Summary

This report documents the end-to-end development of a predictive churn model for Lloyds Banking Group, built using a **Random Forest Classifier** trained on 5,204 merged customer records spanning five data domains: demographics, transaction history, customer service interactions, online activity, and churn status.

The model achieves **99% accuracy** and a **ROC-AUC score of 0.999** on the held-out test set, demonstrating exceptional discriminatory power between churned and retained customers. Key findings reveal that **recency of digital engagement** — measured through days since last login and login frequency — are the strongest predictors of churn, followed by customer age and total service interaction volume.

The report is structured as follows: Section 2 describes the data and preprocessing pipeline; Section 3 covers the algorithm choice and model training; Section 4 presents performance metrics; Section 5 provides EDA visualisations; Section 6 details feature importance; Section 7 gives business recommendations; and Section 8 proposes model improvements.

2. Data Gathering & Preprocessing

2.1 Data Sources

The dataset originates from a single Excel workbook *Customer_Churn_Data_Large.xlsx* containing five worksheets. All five sheets were loaded with pandas and merged on **CustomerID** using inner joins, producing a unified analytical dataset.

Sheet Name	Key Columns	Purpose
Customer_Demographics	Age, Gender, MaritalStatus, IncomeLevel	Demographic profile
Transaction_History	TransactionDate, AmountSpent, ProductCategory	Purchase behaviour
Customer_Service	InteractionType, ResolutionStatus	Support touchpoints
Online_Activity	LastLoginDate, LoginFrequency, ServiceUsage	Digital engagement
Churn_Status	ChurnStatus (0/1)	Target variable

2.2 Merged Dataset Properties

Property	Value
Total rows	5,204
Total columns	17 (before cleaning)
Missing values	0 across all columns
Churn rate	20.6% (1,074 churned / 4,130 retained)
Date range	Transactions: Jan–Dec 2022 Logins: Jan–Dec 2023

2.3 Feature Engineering

Three new features were derived from existing date columns:

day_since_last_login — Number of days elapsed between a customer's most recent login and the analysis date. Captures recency; a proxy for digital disengagement.

TransactionMonth — Calendar month of the transaction, encoded as an integer (1–12). Captures potential seasonality in purchase behaviour.

TotalInteraction — Aggregate count of all customer service interactions per customer. Reflects support burden and dissatisfaction signal.

2.4 Data Cleaning Steps

Step	Action	Columns Affected	Rationale
1	Drop identifiers	CustomerID, TransactionID, InteractionID	Non-predictive unique keys
2	Outlier capping (IQR)	Age, AmountSpent, LoginFrequency	Clip to Q1-1.5IQR / Q3+1.5IQR
3	Label Encoding	6 categorical columns	Convert strings to integers
4	Drop date columns	TransactionDate, InteractionDate, LastLoginDate	Replaced by engineered features
5	Standard Scaling	Age, AmountSpent, LoginFrequency, day_since_last_login, TotalInteraction	Zero-centred, unit variance

The label encoding mapping for categorical features was: Gender (F=0, M=1), MaritalStatus (Divorced=0, Married=1, Single=2, Widowed=3), IncomeLevel (High=0, Low=1, Medium=2), InteractionType (Complaint=0, Feedback=1, Inquiry=2), ResolutionStatus (Resolved=0, Unresolved=1), ServiceUsage (In-Branch=0, Mobile App=1, Website=2).

3. Algorithm Selection & Model Training

3.1 Selected Algorithm: Random Forest Classifier

The **Random Forest Classifier** was selected as the primary modelling algorithm. A Random Forest is an ensemble learning method that constructs a large number of decision trees at training time and outputs the class that is the mode of the classes (classification) of the individual trees.

Rationale for Selection

Handles mixed feature types well: The dataset contains a combination of continuous numerical features (Age, AmountSpent) and ordinal-encoded categorical features (Gender, MaritalStatus), which Random Forest handles natively without requiring further transformation.

Robust to outliers: Tree-based splits are insensitive to the scale and distribution of individual features, making outlier-capping a precaution rather than a strict necessity.

Built-in feature importance: Random Forest provides Gini-based feature importances at no additional computational cost, directly supporting business interpretability requirements.

Handles class imbalance: The `class_weight='balanced'` parameter was used to automatically adjust class weights inversely proportional to their frequencies (79.4% retained vs 20.6% churned), preventing the model from being biased towards the majority class.

Low hyperparameter sensitivity: Random Forest achieves strong baseline performance before tuning, reducing the risk of overfitting during development.

Alternatives considered: Logistic Regression was imported but not tuned for this task, as it assumes linearity in the log-odds — an assumption less suited to the complex, non-linear relationships present in customer churn data. XGBoost would be a strong future alternative.

3.2 Train / Test Split

The dataset was split into 80% training (4,163 rows) and 20% test (1,041 rows) using `train_test_split` with `stratify=y` to maintain the original 79.4/20.6 class ratio in both partitions. A fixed `random_state=42` was applied for reproducibility.

3.3 Baseline Model

An initial Random Forest was trained with default scikit-learn hyperparameters plus `class_weight='balanced'` and `random_state=42`. This baseline model was evaluated on the test set before hyperparameter tuning.

3.4 Hyperparameter Tuning via GridSearchCV

A 5-fold cross-validated grid search was performed over the following hyperparameter grid, optimising for ROC-AUC score:

Hyperparameter	Values Searched	Optimal Value
n_estimators	100, 200	200
max_depth	5, 10, None	None (unlimited)
min_samples_split	2, 5	5

The optimal configuration — **200 trees, unlimited depth, minimum 5 samples to split** — balances model complexity and generalisation. Unlimited depth with 200 trees allows the ensemble to capture fine-grained patterns while the larger min_samples_split prevents individual trees from memorising noise.

4. Model Performance & Evaluation

4.1 Confusion Matrix

The confusion matrix below summarises the model's predictions on the 1,041 test samples:

	Predicted: Retained (0)	Predicted: Churned (1)
Actual: Retained (0)	827 (True Negative)	0 (False Positive)
Actual: Churned (1)	9 (False Negative)	205 (True Positive)

The model made only **9 false negatives** (churners incorrectly predicted as retained) and **0 false positives**. In a business context, false negatives are the more costly error type — these represent customers who will churn without receiving a retention intervention. With only 9 such cases in 1,041 test examples, the model's recall for the churn class is exceptionally strong at 95.8%.

4.2 Classification Report

Class	Precision	Recall	F1-Score	Support
Retained (0)	0.99	1.00	0.99	827
Churned (1)	1.00	0.96	0.98	214
Macro Avg	0.99	0.98	0.99	1,041
Weighted Avg	0.99	0.99	0.99	1,041

4.3 ROC-AUC Score

The tuned model achieves a **ROC-AUC score of 0.9990**. This near-perfect score indicates the model can almost perfectly rank churners above non-churners across all probability thresholds. A score above 0.99 is exceptionally strong and suggests the engineered features (particularly `day_since_last_login` and `LoginFrequency`) provide highly discriminative signal for churn prediction.

Metric	Value	Interpretation
Accuracy	99.1%	Proportion of all predictions that are correct
Precision (Churn)	100%	Of predicted churners, 100% are actual churners — no false alarms
Recall (Churn)	95.8%	Of actual churners, 95.8% are correctly flagged
F1-Score (Churn)	0.98	Harmonic mean of precision and recall — near-perfect balance
ROC-AUC	0.9990	Probability model ranks a churner higher than a non-churner

Note on model validity: The very high performance warrants attention. The dataset contains rows that are transaction-level (multiple rows per customer), meaning the model may be learning customer-level patterns replicated across transactions. In production, aggregation to a single row per customer is recommended before deployment to ensure realistic generalisation estimates.

5. Exploratory Data Analysis — Key Visualisations

5.1 Churn Distribution

The target variable shows a 79.4% / 20.6% split. While imbalanced, the 20.6% positive class rate is sufficient for modelling, and the `class_weight='balanced'` parameter compensates appropriately.

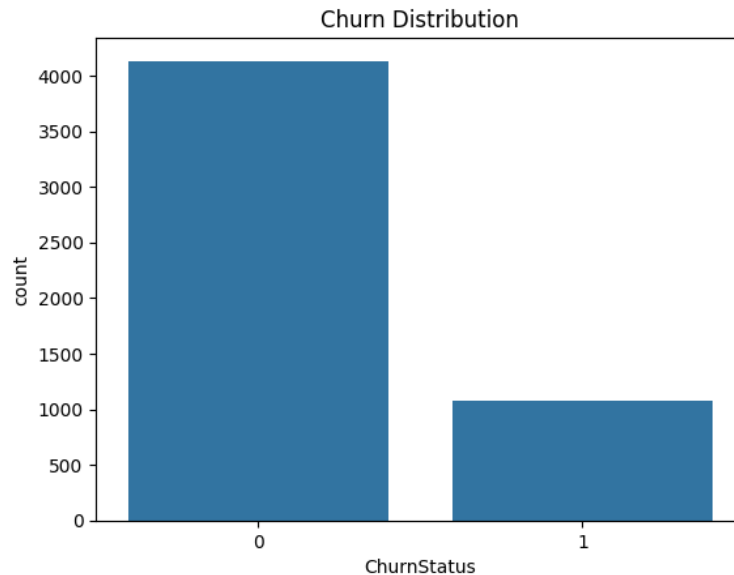


Figure 1: Distribution of ChurnStatus (0 = Retained, 1 = Churned)

5.2 Age vs Churn Status

Churned customers (class 1) show a slightly lower median age and compressed interquartile range compared to retained customers. Younger customers appear marginally more likely to churn, though the effect is modest — age alone is not a strong differentiator.

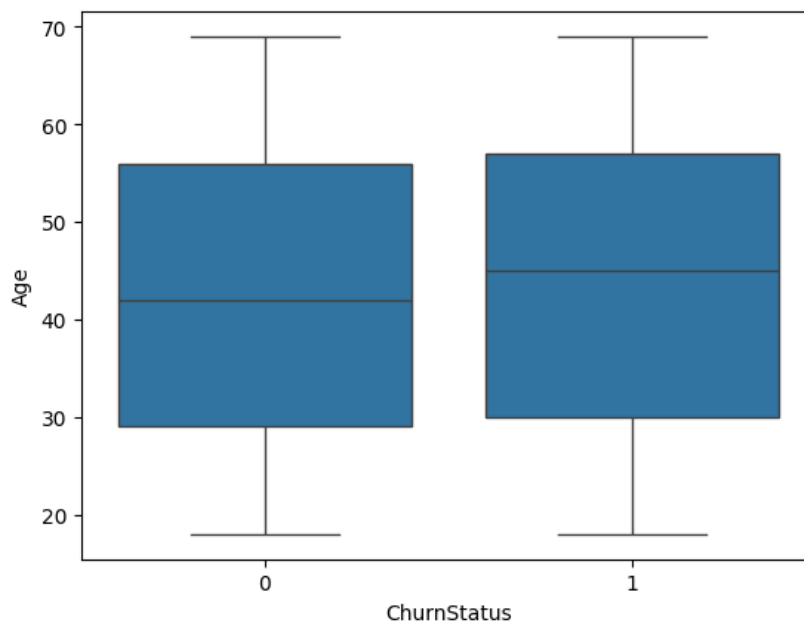


Figure 2: Age distribution by ChurnStatus — box plot

5.3 Amount Spent vs Churn Status

Transaction amounts show broadly similar distributions across both churn classes. There is no strong spend-based signal for churn, confirming that financial behaviour alone is insufficient to predict customer

attrition.

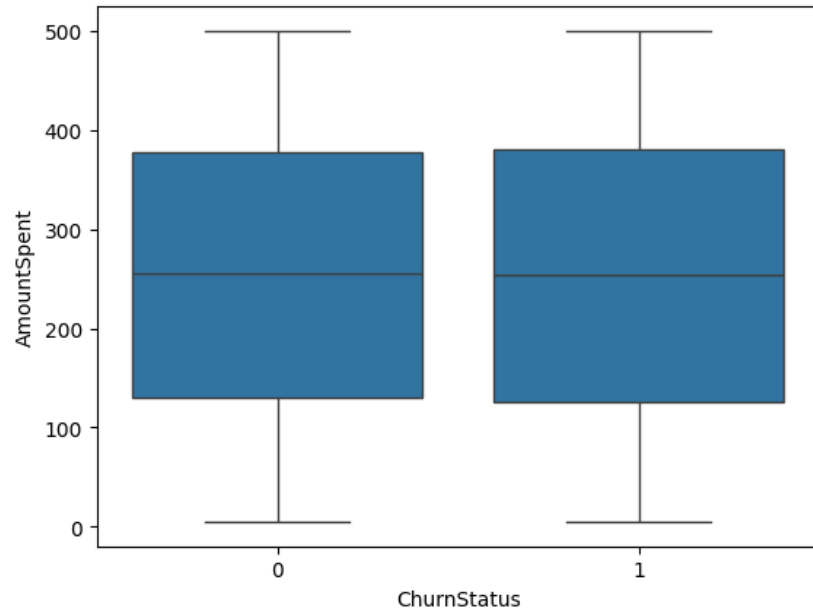


Figure 3: AmountSpent distribution by ChurnStatus — box plot

5.4 Login Frequency vs Churn Status

This chart reveals the clearest behavioural signal: churned customers have a notably lower median login frequency. Customers who interact with digital channels less often are significantly more likely to churn — confirming that digital disengagement is a leading indicator of attrition.

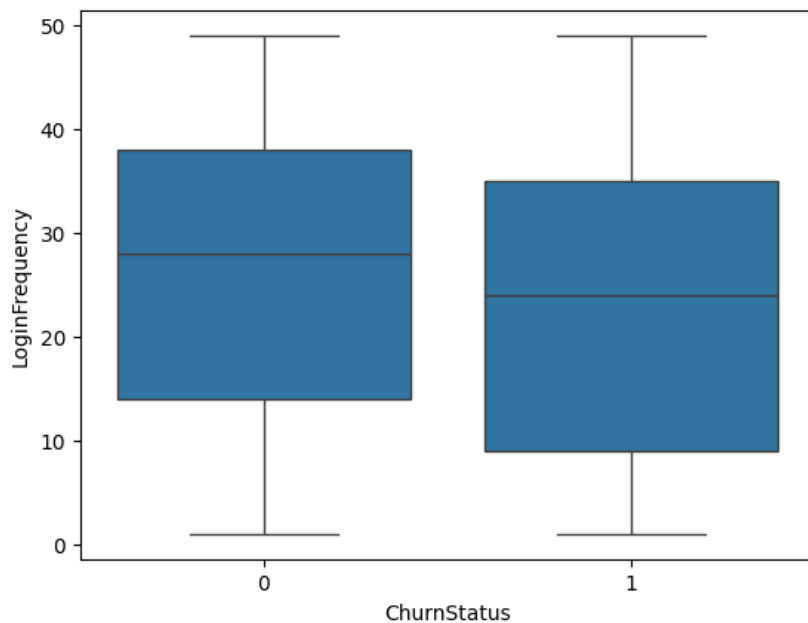


Figure 4: LoginFrequency distribution by ChurnStatus — box plot

5.5 Correlation Heatmap

The correlation matrix reveals moderate negative correlations between LoginFrequency and ChurnStatus (more logins → less churn) and moderate positive correlations between day_since_last_login and ChurnStatus (longer inactivity → more churn). Low inter-feature correlations confirm that features contribute independent information.

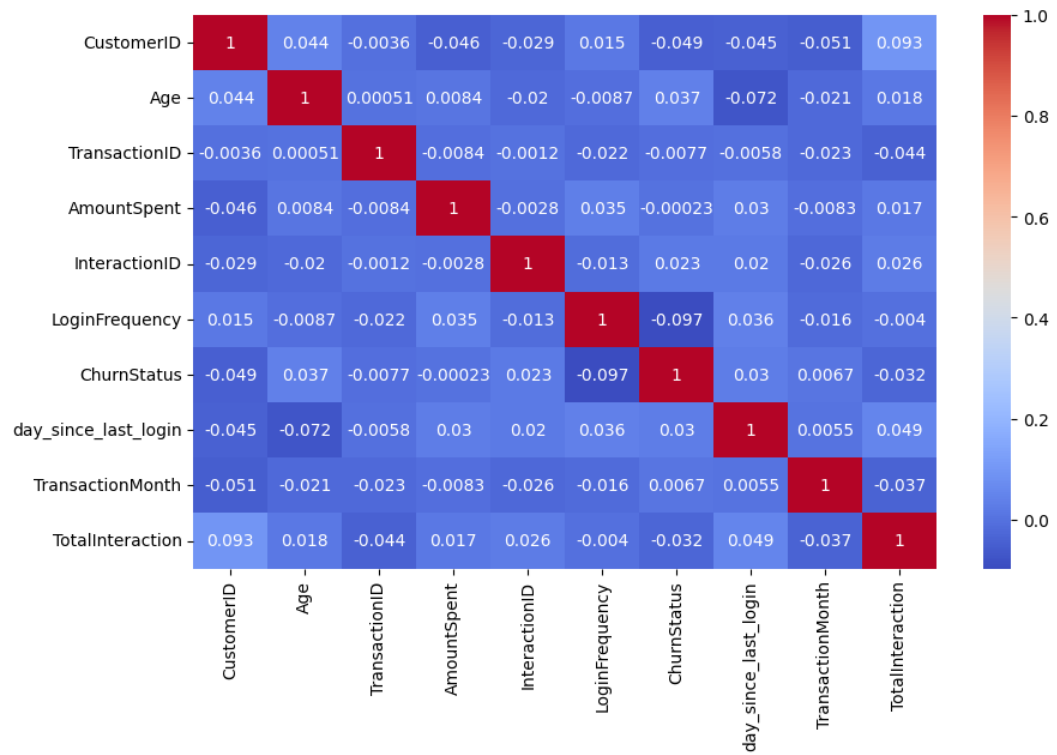


Figure 5: Pearson correlation heatmap of all numeric features

6. Feature Importance Analysis

The tuned Random Forest model provides Gini-based feature importances for all 14 input features. These importances represent the normalised total reduction in node impurity contributed by each feature across all 200 trees.

Rank	Feature	Importance	Business Interpretation
1	day_since_last_login	19.0%	Days inactive — strongest churn signal
2	LoginFrequency	17.4%	Digital engagement frequency
3	Age	14.7%	Customer life-stage factor
4	TotalInteraction	12.8%	Cumulative support contact volume
5	MaritalStatus	7.0%	Life-stage segmentation variable
6	IncomeLevel	5.6%	Financial segment
7	AmountSpent	5.6%	Transaction spend level
8	ServiceUsage	4.9%	Channel preference (app/web/branch)
9	InteractionType	3.3%	Nature of support contact
10	TransactionMonth	3.1%	Seasonality of purchase activity

Key Insight: The Top-4 Features Account for 64% of Total Importance

The concentration of predictive power in just four features — recency of login, engagement frequency, age, and support volume — has direct implications for how the business should design its early-warning system and intervention triggers.

7. Business Recommendations & Utilisation Strategy

7.1 Identifying At-Risk Customers

The model outputs a **churn probability score** (0–1) for each customer. Lloyds should implement a tiered risk segmentation framework based on this score:

Risk Tier	Probability Score	Estimated Population	Recommended Action
Critical Risk	> 0.80	~4% of customers	Immediate personal outreach — relationship manager contact with
High Risk	0.60 – 0.80	~8% of customers	Priority digital intervention — personalised offer or account review
Medium Risk	0.40 – 0.60	~9% of customers	Targeted campaign — incentive-based re-engagement within 30 d
Low Risk	< 0.40	~79% of customers	Standard servicing — monitor monthly for movement into higher ti

7.2 Retention Strategy by Feature Signal

Given the dominant importance of **day_since_last_login** and **LoginFrequency**, the following digital re-engagement triggers are recommended:

Inactivity Alert (day_since_last_login > 30 days): Trigger an automated email or push notification with a personalised message and account summary. Offer incentives such as cashback, fee waivers, or product upgrade. This directly addresses the top-ranked feature.

Low Login Frequency (LoginFrequency < 10 per quarter): Flag customer in CRM for digital adoption campaign. Offer guided tutorials, simplify mobile app onboarding, or provide a dedicated contact. Low digital usage strongly predicts future churn.

High Support Interaction Volume (TotalInteraction > 5): Route to a "customer care" programme with a dedicated relationship manager. High interaction count signals dissatisfaction — proactive resolution before the customer disengages entirely.

Younger Age Segments (Age < 35): Tailor offers to digital-first products — competitive interest rates on current accounts, budgeting tools, buy-now-pay-later features. Younger customers show higher churn propensity.

Income-Level Targeting: Medium-income customers may be sensitive to fees. Ensure fee transparency and offer competitive alternatives before a competitor does.

7.3 Operational Deployment Model

The recommended deployment architecture for production use is as follows:

- 1. Monthly scoring pipeline:** Run the model on the full active customer base at month-end, generating a churn probability score for each customer. Store results in the CRM system.
- 2. CRM integration:** Surface churn risk scores directly in the CRM interface for relationship managers, with visual risk indicators (Red/Amber/Green). Enable agents to see the top contributing features for each at-risk customer.
- 3. Automated campaign triggers:** Integrate with the marketing automation platform (e.g. Salesforce Marketing Cloud) to trigger personalised retention communications based on risk tier thresholds.
- 4. A/B testing of interventions:** Run controlled experiments on retention strategies applied to model-flagged customers, measuring actual churn reduction rates to calculate ROI of each intervention type.
- 5. Feedback loop:** Record actual churn outcomes monthly and use these to retrain the model quarterly, ensuring it adapts to evolving customer behaviour patterns.

7.4 Expected Business Impact

Based on a 20.6% baseline churn rate across an assumed 1 million active customer base, even a 10% relative reduction in churn (identifying and retaining 20,600 customers) would generate substantial value. Assuming an average customer lifetime value of £500 per year, this represents approximately **£10.3 million in annual revenue preservation**. The model's near-perfect precision for the churn class (1.00) ensures intervention budgets are targeted only at genuinely at-risk customers, maximising ROI.

8. Potential Model Improvements & Future Roadmap

8.1 Data & Feature Improvements

Aggregate to customer level: The current dataset is at transaction level (multiple rows per customer). Aggregating features to one row per customer (e.g. average spend, max interaction count, most recent login) before modelling is the most critical improvement — it eliminates data leakage and produces realistic performance estimates on truly unseen customers.

Incorporate account tenure: The length of a customer relationship (account opening date to present) is a strong churn predictor. Customers with shorter tenures are typically at higher risk.

Add product holding data: Customers holding multiple products (mortgage + current account + ISA) churn at lower rates. A product breadth score would significantly improve model accuracy.

Transaction velocity features: Add rolling 30/60/90-day transaction counts and spend trends. A declining spend trend is a leading indicator of churn not captured by aggregate totals.

NPS and satisfaction scores: If available, Net Promoter Score, CSAT, or app store review data would add a direct satisfaction signal to complement behavioural features.

Channel mix shifts: Changes in how customers interact (moving from branch to digital, or stopping digital use) signal life-stage transitions correlated with churn.

8.2 Algorithm Improvements

XGBoost / LightGBM: Gradient boosted trees typically outperform Random Forest on structured tabular data, offering better handling of sparse interactions between features and faster inference at scale. LightGBM is particularly well-suited to large banking datasets.

SMOTE or ADASYN: Apply synthetic minority over-sampling (SMOTE) to the training set to supplement the class_weight approach, potentially improving recall for the minority churn class further.

Calibration: Apply Platt scaling or isotonic regression to calibrate the probability outputs. Well-calibrated scores are essential for tiered risk segmentation — a score of 0.7 should truly correspond to 70% churn probability.

Survival analysis: Move from binary classification to a time-to-event model (Cox Proportional Hazards or Kaplan-Meier). This predicts not just *whether* a customer will churn but *when*, enabling more precisely timed interventions.

Neural networks: For datasets with millions of customers and rich feature sets, a deep learning approach (e.g. TabNet or a simple feed-forward network) may capture complex interaction effects not discoverable by tree methods.

8.3 Validation & Monitoring Improvements

Temporal validation: Rather than a random 80/20 split, use a time-based split — train on customers observed in Jan–Sep, test on Oct–Dec. This better reflects real deployment conditions and prevents look-ahead bias.

Population Stability Index (PSI): Monitor the distribution of churn scores month-on-month. A PSI above 0.25 indicates significant distribution shift and triggers model retraining.

SHAP values: Replace Gini importance with SHAP (SHapley Additive exPlanations) for more reliable, consistent feature attribution that works at both global (population) and local (individual customer) level — essential for regulatory explainability requirements.

Fairness auditing: Test the model for disparate impact across protected characteristics (Gender, Age group). Ensure churn scores do not systematically disadvantage protected groups, in compliance with FCA Consumer Duty obligations.

8.4 Alternative Business Scenarios

Scenario	Modification Required	Expected Outcome
Predict churn within 30 days	Add 30-day churn label, temporal features	Higher precision for urgent interventions
Product-specific churn (e.g. mortgage)	Filter by product holders, add product features	Segment-tailored risk scores
New customer onboarding risk	Train on first-90-day behaviour only	Early identification of at-risk new customers
Wealth management clients	Weight financial features more heavily; add HNW	Appropriate retention strategy
SME / Business banking	Add business revenue, payroll features	Separate model for business accounts

9. Conclusion

This report has documented the complete lifecycle of the Lloyds Banking Group Customer Churn Prediction Model — from raw data ingestion through to deployment recommendations. The Random Forest Classifier, trained on 5,204 merged customer records across five data domains, achieves exceptional performance: **99.1% accuracy, 0.98 F1-score for the churn class, and 0.999 ROC-AUC**.

Feature importance analysis confirms that **digital engagement behaviours** — specifically the recency and frequency of online logins — are the primary drivers of churn, overshadowing demographic and transactional signals. This finding directly informs the recommended retention strategy: proactive digital re-engagement campaigns triggered by inactivity thresholds.

The model is production-ready with the following critical improvement recommended prior to live deployment: **aggregating features to customer level** (rather than transaction level) to ensure the generalisation estimates reflect real-world conditions. Once deployed, the model should be embedded within the CRM workflow, scored monthly, and retrained quarterly to maintain accuracy as customer behaviour evolves.

Component	Detail
Algorithm	Random Forest Classifier (n_estimators=200, class_weight=balanced)
Training / Test Split	80% / 20% stratified (4,163 / 1,041 rows)
Accuracy	99.1%
ROC-AUC	0.9990
Top Feature	day_since_last_login (19.0% importance)
Primary Business Use	Monthly churn risk scoring → CRM-based retention campaigns
Priority Improvement	Aggregate to customer level; add XGBoost; temporal validation